

Tyler Lamme

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Security Clearance: Active DoD Secret Clearance

EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

Bachelor of Science in Computer and Systems Engineering (Computer Engineering)

Sept. 2023 – May 2027

- GPA: 3.82 | Dean's Honor List (All Semesters)
- Relevant Coursework: Data Structures, Algorithms, Foundations of CS, Computer Architecture, Computer Networks, Network Security, Embedded Control, Computational Vision, Signals & Systems, Quantum Devices
- Clubs: RPai (AI Club), RPISEC (Cybersecurity Club), Quantum Computing Club

WORK EXPERIENCE

Raytheon (RTX)

Tewksbury, MA

Radar Systems Engineering Intern (LTAMDS), Modeling and Simulation

May 2026 – Aug. 2026

- Debugged MATLAB codebase by identifying incorrect variables within `.mat` data structures through exploratory plotting and validation, correcting defects and documenting fixes for future analysis.
- Developed a VBA automation tool integrated with Excel and Word to generate PDF deliverables from 100+ records, reducing over 6 hours of manual processing per run while eliminating copy errors.
- Validated software updates through structured testing and created technical documentation detailing data structure corrections and workflow improvements for engineering teams.

Scientific Research Corporation

Huntsville, AL

Radar Engineer Intern, Waveform Generation

May 2025 – Aug. 2025

- Developed Python automation tools to communicate with SCPI-controlled signal analyzers, streamlining RF measurement procedures and simplifying waveform testing procedures for engineers.
- Created a Python-based waveform conversion interface that eliminated manual waveform regeneration when transitioning between software-defined radio (SDR) file formats, improving flexibility of testing workflows.
- Collaborated with engineers to validate waveform generation and SDR performance using laboratory signal analysis equipment, documenting tools and procedures for future users.

Advanced Technologies and Services, Inc.

Wyckoff, NJ

Software Engineer Intern

May 2024 – Aug. 2024

- Automated IoT device production using Arduino firmware and Linux Bash scripts, reducing device preparation time by over 50% while minimizing manual setup errors and improving yields.
- Developed Python-based TCP/UDP network testing tools using Python and Linux sockets and deployed testing frameworks on Google Cloud infrastructure to evaluate distributed IoT device performance.
- Contributed to a web-based ISP network management platform using Angular, FastAPI, and cloud services, supporting backend API development and device communication workflows.

PROJECTS

Dynamic Geospatial Object Search | C++, Python, Embeddings, Vector Search, VLMs

- Developing a C++/Python system to dynamically search satellite imagery for objects specified by text or reference images using embedding models, vector similarity search, and vision-language models.
- Designing a highly optimized multi-stage retrieval pipeline using precomputed image embeddings, nearest-neighbor search, VLM filtering and grounding, and on-demand imagery retrieval to minimize storage and compute.

TylerGPT (AI Portfolio Agent) | Python, FastAPI, React, OpenAI API

- Developed and deployed a full-stack AI portfolio assistant using React and FastAPI with OpenAI tool calling, enabling automated contact capture and unanswered-question tracking.
- Engineered an asynchronous LLM streaming backend with recursive tool execution, conversational context management, REST APIs, Docker deployment, and automated GitHub CI/CD across Vercel and Railway.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: React, FastAPI, Angular, PyTorch, Hugging Face, OpenCV, NumPy, Pandas, Matplotlib

AI/ML: Deep Learning, Object Detection & Segmentation (SAM), LLM Tool Calling, Prompt Engineering

Tools & Embedded: Git, Linux, Docker, REST APIs, MATLAB, Arduino, Ethernet, SPI, PWM, Jupyter Notebooks